

From Representation to Realization: Stage-Resolved Causal Attribution in Apparently Agentic Transitions

Abstract

Apparently agentic outcomes can support multiple causal organizations and therefore underdetermine focal causal attribution. We present a stage-resolved causal-attribution protocol that separates candidate participation, selection provenance, and realization provenance. The protocol uses inherited structural-causal, mediation, transportability, causal-abstraction, representation-intervention, action-selection, and execution machinery rather than proposing a new causal formalism or action mechanism. Its distinctive requirement is evidential composition: a causal proposition established locally may support a stronger end-to-end attribution only when its prerequisites remain valid and any source-target use has an independently justified validity basis. The protocol preserves nonfocal preparation, bypass, and joint focal-nonfocal contribution rather than converting local focal effects into claims of unrestricted autonomy. It also identifies where stronger attribution must remain unresolved when a required proposition, handoff, or provenance contrast is not established.

Keywords: action selection; causal abstraction; causal attribution; causal inference; causal provenance; system boundaries; systems methodology

Practitioner Points

- An observed outcome alone does not establish whether a represented alternative participated causally in selection, whether a focal subsystem supplied the selected state, or whether it implemented the realized transition.
- Strong end-to-end attribution requires stage-specific causal evidence and an explicit validity basis when prerequisites or causal effects are carried across experimental regimes.
- When a required certificate, provenance contrast, or validity condition is unavailable, the warranted result is a bounded or unidentified attribution rather than inferred focal control.

1. The Attribution Problem

1.1 Apparently agentic behavior underdetermines causal provenance

Consider an observed transition $\mathcal{V}_A$ in a setting where multiple possible courses are representable and the realized outcome corresponds to alternative A . The completed transition may appear sufficient for a simple focal attribution. Causally, however, the same observed result can remain compatible with different organizations upstream of the outcome.

A represented alternative may never causally participate in the dynamics relevant to selection. A participating alternative may be present but lose. A focal state may genuinely contribute to selection while itself having relevant nonfocal provenance. A selected alternative may fail to produce focal implementation, and a successful focal implementation may fail to produce the realized environmental transition. Conversely, nonfocal assistance or bypass may contribute to an outcome that matches the focal alternative. Conventional action science already rejects a unitary treatment of intentional action and distinguishes multiple functional components of action specification and control (Brass & Haggard, 2008; Pacherie, 2008; Chambon et al., 2014), while affordance-competition accounts permit several potential actions to be specified before selection is resolved (Cisek, 2007).

The observed outcome therefore does not by itself identify the causal provenance of representational availability, candidate causal participation, selection, focal implementation, or realization. Similar outcomes can be compatible with different answers to each of those causal questions. The methodological problem is consequently narrower than determining whether a transition should be classified as agentic: **what focal causal attribution does the available evidence actually license?**

The protocol developed below answers that question by requiring each relevant causal proposition to be established at the attribution stage where it is needed and by requiring an explicit validity basis whenever an upstream proposition is carried into a different downstream regime.

1.2 Boundary-, query-, and regime-relative attribution

Let B denote the analytical boundary separating focal from nonfocal variables for the attribution query. The distinction is methodological rather than metaphysical: changing B changes which variables are represented as focal or nonfocal and therefore which causal-provenance questions can be expressed and tested.

Systems research has long treated boundary judgments as methodologically consequential because changing what is included or excluded can change the system description and the scope of inquiry (Ulrich, 1987; Midgley, 2000). Recent SRBS work likewise develops multilevel system-boundary judgments for complex settings (Pinzón-Salcedo & Bernal-Alvarado, 2022). The present use of B is narrower: it is a query-specific causal-attribution boundary separating variables treated as focal from those treated as nonfocal for the causal-provenance question at issue. This is compatible with the general systems insight that boundary choice matters analytically, but it does not import the normative, stakeholder-oriented, or critical-reflexive apparatus of boundary critique. Here, changing B matters only insofar as it changes which causal-provenance contrasts can be represented, tested, or licensed.

Causal claims in this manuscript are accordingly relative to the variables represented within and outside B , the causal query, the observation family, and the intervention or regime under which a proposition is identified. Structural causal models provide the inherited distinction between observational and interventional claims and the modular semantics of intervention (Pearl, 2010). Mediation and path-specific-effect frameworks already provide tools for distinguishing direct, indirect, and path-restricted causal contributions when the graph and

identifying assumptions support those queries (Robins & Greenland, 1992; Avin et al., 2005). Transportability theory provides formal source-target conditions under which causal results learned in one environment may be licensed in another (Pearl & Bareinboim, 2011, 2014). Exact and approximate causal transformations provide related machinery for interventional consistency across causal descriptions and abstraction levels (Rubenstein et al., 2017; Beckers & Halpern, 2019; Beckers et al., 2020; Rischel & Weichwald, 2021; Xia & Bareinboim, 2025).

Boundary enlargement can therefore refine an attribution without erasing a valid local causal relation. If a focal state is interventionally established to affect selection, later identification of a nonfocal cause of that focal state changes its upstream provenance; it does not, by itself, negate the focal state's downstream causal effect. The causal account should preserve both facts where both are supported.

This scope also limits the strongest conclusion available. Tested focal causal contribution can be identified relative to B and the nonfocal variables actually represented and tested. The analysis does not establish the absence of every conceivable causal antecedent outside that boundary. Bounded attribution may be identifiable when the required causal assumptions, observations, interventions, and validity conditions are satisfied; unrestricted causal origin lies outside the licensed claim.

1.3 Contribution and inherited machinery

The protocol does not introduce a new causal calculus, transportability theorem, causal-abstraction formalism, or action architecture. It inherits structural causal models, mediation and path-specific effects, causal transportability, exact and approximate causal transformations, representation-level interventions, ordinary action-selection science, and established distinctions among selection, implementation, and action effects. Actual-causation theory is used only to delimit stronger token-level claims that ordinary interventional contribution does not automatically establish (Halpern, 2016; Pearl, 1999; Li & Pearl, 2023).

The proposed contribution is an **integrated stage-resolved attribution discipline**. The attribution roles may be instantiated by parallel, recurrent, overlapping, distributed, or continuously evolving physical dynamics; their logical order does not imply a universal serial biological pipeline. For the complete end-to-end attribution query considered here, it requires separate certification of candidate causal participation, selection provenance, and realization provenance; an explicit validity basis at the interfaces between those certifications; and preservation of nonfocal preparation, bypass, and joint focal-nonfocal contribution. Local focal causal contribution is therefore not promoted to unrestricted focal autonomy without additional evidence.

Existing causal-inference, causal-abstraction, representation-intervention, and action-control literatures provide the component machinery. The contribution proposed here is the requirement that the appropriate inherited causal query be answered at the correct attribution stage, that the proposition passed downstream be explicitly certified in the regime where it is needed, and that unresolved provenance prevent promotion to a stronger end-to-end focal attribution.

Prior-art positioning.

Prior-art family	What it already supplies	What the present protocol additionally requires
SCM / interventionism	Interventional effects, modularity, graph-based causal queries	Stage-specific attribution propositions and explicit interface certification before end-to-end promotion
Mediation / path-specific effects	Direct, indirect, and selected-path causal effects	Use of those estimands at the appropriate attribution stage without treating path effects as complete end-to-end provenance
Transportability	Formal source-target conditions for carrying causal effects across environments	A handoff record stating which upstream proposition is being carried, from which regime to which downstream regime, and on what inherited validity basis
Causal abstraction / exact transformation	Interventional consistency across model levels or representations	Preservation of the attribution proposition required at each interface, rather than abstraction consistency alone
Interventionally validated representations	Intervention-based evidence that internal representations have causal roles	A separate candidate-participation certificate tied to a validated preselection target before selection provenance is inferred
Action-stage decomposition	Nonunitary intention, selection, execution, and effect structure	Independent causal certification of participation, selection provenance, and realization provenance rather than stage labels inferred from outcome
Empirical action-selection / motor-control paradigms	Parallel potential-action specification, preparation-versus-movement distinctions, and nonserial/recurrent architectures (Cisek & Kalaska, 2005, 2010; Kaufman et al., 2014; Kim et al., 2021)	Uses those empirical roles only when independently validated for the attribution query; does not infer the complete protocol from stage-correlated signals
Causal-neuroscience link validation	Multi-link causal interpretation of	Applies the same inherited validity discipline to

Prior-art family	What it already supplies	What the present protocol additionally requires
	perturbation-to-brain-to-cognition-to-behavior chains, including propagation and state-dependence concerns (Bergmann & Hartwigsen, 2021; Weichwald & Peters, 2021)	candidate participation, selection provenance, realization provenance, and their typed interfaces
Systems boundary judgment / boundary critique	Recognition that system description and inquiry depend on consequential inclusion/exclusion and boundary judgments	A query-specific focal/nonfocal causal-attribution boundary tied to explicit causal-provenance contrasts, intervention regimes, and admissibility of downstream attribution

The protocol is therefore narrower than a theory of agency. It neither defines agency nor identifies a privileged controller. It specifies the evidence required to support increasingly strong focal attributions for an apparently agentic transition whose outcome alone underdetermines its causal provenance.

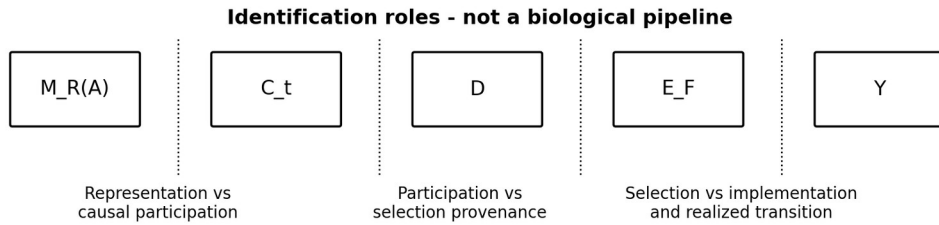


Figure 1. Stage-resolved identification schematic for focal causal attribution. $M_R(A)$ denotes representational availability, C_t the validated candidate-participation set, D the independently identified selected state, E_F the focal implementation outcome, and Y the realized transition. The separators mark three identification failures: representation does not establish candidate causal participation; participation does not establish selection provenance; and selection does not establish focal implementation or realized outcome. This is an **identification schematic, not a proposed serial biological mechanism.**

2. Three Identification Failures

The attribution problem can be decomposed before specifying any particular intervention. The relevant propositions concern whether an alternative is represented, whether it causally participates in the process from which selection emerges, what causally contributes to the selected state, whether the selected state is implemented by the focal subsystem, and whether the corresponding transition is ultimately realized. Existing memory, action-selection, and action-control literatures already permit distinctions among these functions (Stokes, 2015; Cisek, 2007; Brass & Haggard, 2008; Pacherie, 2008; Chambon et al., 2014). The methodological

issue here is that evidence for one proposition cannot automatically substitute for evidence required by another.

2.1 Representation is not causal participation

Let $M_R(A)=1$ denote that alternative A satisfies the representational-availability criterion relevant to the experiment. Such evidence may take several forms: A may be retained, retrievable, reportable, or decodable. None of those observations alone establishes that A currently exerts a causal effect on the dynamics from which selection emerges:

$$\boxed{M_R(A) = 1 \not\Rightarrow A \in \mathcal{C}_t(\mathcal{J}, \mathcal{O}).} \quad (1)$$

Working-memory research provides concrete examples in which maintained information need not map simply onto continuously expressed persistent activity, and perturbational work can reveal or reactivate latent memory content (Stokes, 2015; Rose et al., 2016; Wolff et al., 2017). These results show why representational availability, detectability, and perturbational responsiveness must be distinguished from the particular causal-participation query addressed here; they do not themselves certify participation in selection.

Downstream behavior does not repair the missing attribution. If the realized transition favors incumbent B , the outcome alone cannot determine whether A failed to participate causally or participated and lost. Representation therefore licenses a participation question; it does not answer it.

2.2 Participation is not selection provenance

Suppose A and B have both been certified as causally participating candidates. That result narrows the problem, but it does not identify the provenance of the selected state D . In particular, participation does not establish that the focal subsystem independently supplied the causally effective state from which selection followed.

Selection may contain a focal causal contribution, a nonfocal causal contribution, nonfocal preparation of a causally effective focal state, or joint focal-nonfocal contribution. These contrasts may be represented using ordinary SCM, mediation, or path-specific-effect tools as appropriate to the graph and identifying assumptions. They need not differ in whether A and B genuinely participated beforehand; they differ in the causal provenance of the state that contributed to D .

The relevant distinction is therefore between focal causal effectiveness and the provenance of the causally effective focal state. Participation licenses the selection-provenance query; it does not settle it.

2.3 Selection is not implementation or realization

Selection introduces two further attribution boundaries. Let $D=A$ mean that the domain-specific, independently validated empirical selection criterion for alternative A has been

satisfied. D is an attribution role rather than a universal discrete commitment node: its empirical realization may be distributed, graded, recurrent, population-level, continuously evolving, or time-indexed. That proposition does not by itself establish focal implementation, and focal implementation does not by itself establish realization of the target transition.

When Y extends beyond the focal subsystem, nonfocal conditions can prevent an implemented transition from succeeding or can assist, redirect, or independently produce an outcome compatible with A . Existing work already distinguishes multiple stages or components of intention, action specification, selection, execution, and action effects (Brass & Haggard, 2008; Pacherie, 2008; Chambon et al., 2014). The present issue is inferential: an observed effect cannot retrospectively certify every upstream proposition required to attribute that effect to the focal subsystem.

Representation, candidate causal participation, selection, focal implementation, and realization are therefore distinct attribution propositions, not evidence-interchangeable descriptions of one observed outcome.

3. Local Identification Tests

The protocol separates three local identification problems. Test -1 asks whether a represented alternative participates causally in the dynamics from which selection emerges. For purposes of the complete end-to-end attribution developed here, Test 0 is admissible only after the relevant alternatives have been certified as participating; the local causal relations addressed by Test 0 may nevertheless be studied independently. Likewise, Test 1 is admissible within the complete attribution only after the selected state has been independently established; its local causal relations may also be studied independently. The tests use inherited causal-identification machinery; their role is to prevent evidence obtained at one stage from being promoted into an attribution belonging to another.

3.1 Candidate-participation test

3.1.1 Observational ambiguity

Let $M_A(A)=1$ denote that alternative A satisfies the relevant criterion for representational availability. Representational availability may be behaviorally relevant, but it does not by itself establish current causal participation. The realized incumbent outcome is likewise insufficient:

$$\boxed{Y = B \text{ is compatible with } A \notin \mathcal{C}_t(\mathcal{J}, \mathcal{O}) \text{ and } A \in \mathcal{C}_t(\mathcal{J}, \mathcal{O}), D = B.} \quad (2)$$

In the first case, the represented possibility A does not enter the validated causal dynamics relevant to selection. In the second, A participates but B is selected. Both can terminate in the same realized outcome.

3.1.2 Candidate-participation certificate

The candidate-participation test identifies an **intervention- and observation-relative certificate for the alternative A** , not an untouched-token actual-cause judgment. Let q_A denote the candidate-specific experimentally addressable causal variable or state through which A is tested, and let Z_C denote a candidate-sensitive preselection target. The set $\mathcal{C}_t(\mathcal{J}, \mathcal{O})$ contains alternatives, not experimental variables. Thus the certificate has the form

$$\boxed{A \in \mathcal{C}_t(\mathcal{J}, \mathcal{O})}. \quad (3)$$

q_A supplies evidence for that certificate; it is not silently identified with A .

A certificate for A requires four elements. First, $M_A(A)=1$ must be established by a representational criterion independent of the participation effect being tested. Second, q_A must be candidate-specific enough that its manipulation can be interpreted as addressing A rather than indiscriminately perturbing the selection system. Third, the component of Z_C used for the inference must have **independently established causal relevance to the prespecified selection-attribution query**. Prediction of D , temporal precedence, decoding, attention, salience, arousal, conflict, task engagement, generic motor readiness, or generic network influence is insufficient by itself. This selection-relevance validation must either apply in the regime in which the candidate-participation inference is made or have an independently justified validity or transport basis into that regime. Its validity cannot be defined by the same $q_A \rightarrow Z_C$ effect used to certify participation. Fourth, under the relevant candidate-specific intervention family, variation in q_A must change the validated selection-relevant component of Z_C in the prespecified contrast that counts as evidence of candidate-specific causal influence, while the representational criterion and the qualification conditions required by the local contrast remain satisfied.

This formulation avoids circularity: participation is not defined as whatever makes an arbitrary measured preselection signal change. It is a bounded causal certificate for alternative A relative to the stated causal model, design, identifying assumptions, observation family, and intervention regime. It does not assert that A was a token-level actual cause in an untouched factual run.

Interventionally validating internal representations has close precedent in causal-representation work. For example, interchange interventions have been used to test whether neural-network representations realize the causal roles of aligned high-level variables rather than merely supporting a successful probe (Geiger et al., 2021). The present protocol inherits that interventionist logic but applies it to the narrower candidate-participation attribution.

3.1.3 Intervention limit: detection versus recruitment

The strongest Test -1 interpretation can fail when the intervention changes the participation status whose pre-existing causal role is being queried. Presenting A again, increasing its

salience, or otherwise exciting its representation may recruit or reactivate A even if it did not exert the same causal leverage immediately beforehand. Perturbational working-memory studies make this possibility concrete: external perturbations can expose hidden mnemonic states (Wolff et al., 2017), and transcranial magnetic stimulation can reactivate latent working-memory representations and alter subsequent behavior (Rose et al., 2016).

Those findings do not imply that interventions cannot reveal pre-existing causal participation. They establish the narrower methodological risk: a perturbation may both measure and modify the state of interest. The experiment must therefore distinguish evidence that detects existing causal leverage from evidence that recruits, reactivates, or otherwise creates the leverage being attributed.

A candidate-specific nonconstructive perturbation is conceptually attractive when feasible because it reduces this constructiveness ambiguity. But nonconstructiveness is not sufficient for intervention validity. A perturbation may still fail because of inadequate selectivity or focality, propagation through connected networks, dependence on the ongoing system state, compensatory effects, generic task consequences, construct invalidity, or induction of abnormal/off-manifold states. The protocol does not claim universal technological availability of a valid perturbation and does not equate a nonconstructive probe with a valid candidate-specific causal inference. If no intervention or design can distinguish pre-existing participation from intervention-induced participation while satisfying the relevant construct-validity conditions, the stronger candidate-participation attribution remains unidentified.

3.2 Selection-provenance test

For purposes of the complete end-to-end attribution, Test 0 is admissible only after the relevant candidates have passed Test -1. The local causal relations addressed by Test 0 may nevertheless be studied independently. For the focal contrast considered here, the required starting proposition is therefore $A, B \in \mathcal{Ct}(\mathcal{I}, \mathcal{O})$ under the regime relevant to the selection-provenance attribution.

The selected state D must be identified independently of final realization. Observing an outcome compatible with A cannot by itself establish $D=A$, because later implementation, nonfocal assistance, bypass, or environmental dynamics may generate the same outcome. Test 0 addresses only the provenance of the independently identified selected state.

3.2.1 Illustrative provenance contrasts

Let z denote a focal state hypothesized to contribute to D , and x a relevant nonfocal state. In the global notation used later, z is denoted z_D , and the corresponding nonfocal selection-stage variable x is denoted x_0 . Figure 2 shows illustrative causal contrasts relevant to the attribution query: focal contribution $z \rightarrow D$; nonfocal contribution $x \rightarrow D$; nonfocal preparation of a causally effective focal state

$$\boxed{x \rightarrow z \rightarrow D;} \quad (4)$$

and joint focal-nonfocal contribution, in which both $z \rightarrow D$ and $x \rightarrow D$ are identified. These graphs are not mutually exclusive or exhaustive mechanism classes. A richer graph may contain several of these relations simultaneously.

The local causal contrasts may be implemented with inherited estimands appropriate to the graph and assumptions. Direct or controlled effects may address one contrast; mediation or path-specific effects may address another; edge or path interventions may be useful when the attribution targets a specified causal route (Robins & Greenland, 1992; Avin et al., 2005). The protocol adds no new estimand. Its requirement is that the appropriate inherited causal query be answered at the selection-provenance stage rather than inferred from candidate participation or final outcome.

Two incoming causal arrows do not by themselves establish interaction, synergy, or effect modification. The default term in this manuscript is therefore **joint focal-nonfocal contribution**. Interaction language is reserved for cases in which an appropriate interaction estimand has actually been identified.

If cases matched on the proposed measured present state respond differently under the intervention family relevant to Test 0, the warranted conclusion is only that the present-state description is insufficient for that query. A further present state $[m]$ may be required. If an adequate present state absorbs the relevant historical dependence, raw history is not retained as an irreducible mechanism.

3.2.2 Nonfocal preparation without causal erasure

Equation (4) carries the central provenance distinction. Suppose intervention on z has independently established that changing z changes D . The local relation $z \rightarrow D$ is then supported within the tested regime. Subsequent evidence that nonfocal variable x contributes causally to z does not revoke that local relation. It refines the provenance account to $x \rightarrow z \rightarrow D$.

Two causal facts must therefore be retained simultaneously: z is causally effective with respect to D , and the causally effective state z has relevant upstream provenance in x . Treating discovery of $x \rightarrow z$ as evidence that z did not contribute would erase a valid local effect. Treating $z \rightarrow D$ as evidence that the focal subsystem independently originated z would erase upstream provenance.

Exact transformations, causal abstraction, and compositional abstraction already provide inherited formalisms for preserving relevant interventional relationships across descriptions (Rubenstein et al., 2017; Beckers & Halpern, 2019; Rischel & Weichwald, 2021). The present protocol does not replace those formalisms. It requires that the attribution record retain both the downstream focal effect and its identified upstream preparation when both are supported.

3.2.3 Handoff validity into Test 0

A Test-0 result is interpretable as selection provenance among already-participating candidates only where the candidate-participation proposition required by Test 0 remains valid in the

regime in which the Test-0 claim is used. If a Test-0 manipulation removes A from the validated candidate set, that experimental condition no longer answers the intended selection-provenance question among the same candidates. The condition may remain scientifically informative, but the missing premise cannot be repaired by the downstream result.

Section 4 formalizes this requirement as a certified handoff.

3.3 Realization-provenance test

For purposes of the complete end-to-end attribution, Test 1 is admissible only after $D=A$ has been independently identified in the target attribution regime. The local causal relations addressed by Test 1 may nevertheless be studied independently. Test 1 asks how the selected state relates to focal implementation and how focal and nonfocal causes relate to the scientifically defined realized transition Y .

3.3.1 Selection versus focal implementation

E_F denotes **an independently validated focal implementation outcome that separates focal implementation from downstream realization for the target query**. What qualifies as implementation is domain-specific. E_F is therefore a query role, not a universal mediator or a claim that every system contains a unique implementation state.

The first distinction is

$$\boxed{D = A \not\Rightarrow E_F = A.} \quad (5)$$

Here $E_F=A$ is shorthand for E_F satisfying the prespecified empirical criterion for focal implementation of alternative A . Selection can therefore be correctly identified even when focal implementation fails.

A causal relation $D \rightarrow E_F$ requires intervention, exogenous variation, or another inherited identification strategy sufficient for the target causal query. Where no valid identification of $D \rightarrow E_F$ is available, association between selection and implementation does not supply that relation.

If the scientific outcome Y is defined at the same boundary and level as focal implementation, E_F and Y may validly merge and the implementation-versus-realization distinction disappears from that narrowed query. If implementation and realization remain scientifically distinct, an independent implementation observable remains required.

3.3.2 Focal implementation versus realized outcome

When Y extends beyond the focal implementation boundary,

$$\boxed{E_F = A \not\Rightarrow Y = A.} \quad (6)$$

Nonfocal conditions can prevent, enable, redirect, or independently produce the realized transition. A correctly implemented focal process may fail to generate its corresponding environmental effect; conversely, a nonfocal process may generate an outcome consistent with A even when focal implementation is absent.

3.3.3 Nonfocal preparation, bypass, and joint realization contribution

Figure 2 summarizes illustrative realization contrasts. In the focal path, D contributes causally to E_F which contributes to Y . A nonfocal bypass exists when a nonfocal state contributes to Y without the focal implementation route required by the attribution. Nonfocal preparation occurs when a nonfocal state contributes to E_F after which E_F genuinely contributes to Y . Joint realization contribution occurs when both E_F and a nonfocal variable contribute causally to Y .

These contrasts are not new mechanism categories. Depending on the graph and assumptions, ordinary direct, indirect, path-specific, or edge-intervention estimands may be used to identify them. Interaction or synergy is not inferred merely because both focal and nonfocal arrows enter the same target.

As in Test 0, identifying nonfocal preparation does not erase a valid local focal effect. If $E_F \rightarrow Y$ is established and nonfocal preparation of E_F is later identified, both facts remain part of realization provenance.

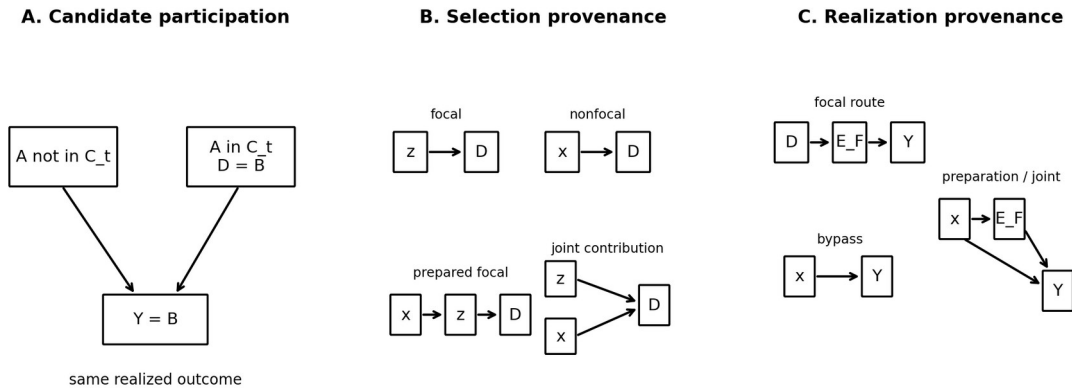


Figure 2. Illustrative local causal contrasts across candidate participation, selection provenance, and realization provenance. Panel A shows why representational availability and the same incumbent outcome are compatible with either nonparticipation of A or participation followed by selection of B . Panel B shows causal contrasts relevant to selection provenance: focal contribution, nonfocal contribution, nonfocal preparation of a causally effective focal state, and joint focal-nonfocal contribution. Panel C shows analogous realization contrasts. These are **illustrative provenance contrasts, not mutually exclusive or exhaustive mechanism classes**; richer causal graphs may contain several simultaneously.

4. Composition and Provenance

The three local tests answer different causal questions. Their end-to-end use therefore requires more than placing locally correct results next to one another. The object being composed is a set of certified causal propositions, and each interface must state why the upstream proposition remains usable in the downstream regime.

4.1 Certified causal propositions and composition

For the complete attribution query, the objects being composed are causal propositions rather than raw state values.

$$\boxed{P_{-1} = [A, B \in \mathcal{C}_t(\mathcal{J}, \mathcal{O})]}. \quad (7a)$$

$$\boxed{P_0 = [D = A; \Pi_D]}, \quad (7b)$$

where Π_D records the identified selection provenance, including the focal, nonfocal, prepared-focal, or joint focal-nonfocal relations supported by the analysis.

$$\boxed{P_1 = [D \rightarrow E_F; \Pi_Y(E_F, x_1, Y)]}, \quad (7c)$$

where Π_Y records the identified realization provenance.

For the complete end-to-end attribution, the admissibility relation is

$$\boxed{P_{-1} \xRightarrow{H_{-1,0}^{\text{pre}}} P_0 \xRightarrow{H_{0,1}^{\text{pre}}} P_1}. \quad (7d)$$

The double arrows in Equation (7d) are **inter-stage prerequisite handoffs**, not biological causal edges and not source-target transport of every causal relation inside the downstream certificate. $H_{-1,0}^{\text{pre}}$ carries the candidate-participation premise required for the selection-provenance attribution. $H_{0,1}^{\text{pre}}$ carries the independently validated selection criterion $D=A$, together with any required identity or mapping of D , into the realization-provenance query. The complete selection-provenance record Π_D remains part of the global causal account even when only a subset of it is required as the local prerequisite for identifying $D \rightarrow E_F$.

A “certificate” means warranted relative to the stated causal model, design, identifying assumptions, observation family, and intervention regime. It does not mean assumption-free causal proof.

The logical ordering

$$\boxed{P_{-1} \prec P_0 \prec P_1} \quad (7e)$$

is an **admissibility order for the manuscript’s complete end-to-end attribution claim**. It is not a claim that local causal relations cannot be studied independently or that experiments must occur chronologically in that order. A single experiment, simultaneous measurements, archival data, a factorial design, or differently ordered studies may establish multiple propositions. What cannot occur is promotion into the complete attribution while a required upstream attribution proposition remains unsupported.

4.2 Inter-stage prerequisite handoffs and within-stage validity

The generic validity record is

$$H = (P, R_s, R_t, V), \quad (8)$$

where P is the proposition being carried, R_s is the source regime in which P was established, R_t is the regime in which P is required, and V is the inherited validity basis for carrying it. A proposition is admissible in R_t either because it is independently revalidated there or because standard causal machinery supplies a justified invariance, transportability, exact-transformation, or abstraction basis. Same-SCM surgical intervention may rely on the modularity assumptions already specified by that SCM; genuine cross-population, cross-measurement, cross-regime, or cross-abstraction use may require a stronger transport argument (Pearl & Bareinboim, 2011, 2014; Rubenstein et al., 2017; Beckers & Halpern, 2019; Rischel & Weichwald, 2021; Xia & Bareinboim, 2025).

The generic record has two distinct uses. $H_{-1,0}^{\text{pre}}$ and $H_{0,1}^{\text{pre}}$ are **inter-stage prerequisite handoffs**. By contrast, a causal proposition belonging within one stage can itself be estimated in a source regime and require separate source-target validation. That operation receives its own typed record and must not borrow an inter-stage label merely because it occurs near a stage interface.

For example, if $D \rightarrow E_F$ is established in a local realization-test source regime R_{1s} and used in target regime R_{1t} its validity is represented separately as

$$H_{1,D \rightarrow E_F}^{\text{val}} = (D \rightarrow E_F, R_{1s}, R_{1t}, V_1). \quad (9)$$

Equation (9) carries only the $D \rightarrow E_F$ proposition. It is not $H_{0,1}^{\text{pre}}$, and its validity does not establish the separate realization-provenance relation $E_F \rightarrow Y$ or any nonfocal realization alternatives.

4.3 Intervention on the selected state and a worked counterexample

Under standard structural intervention semantics, a literal $d\alpha(D=A)$ replaces the ordinary structural determination of D with the imposed value A . Incoming causal arrows to D are therefore severed in the interventional model (Pearl, 2010). An experiment estimating $D \rightarrow E_F$

under such an intervention cannot simultaneously be described as preserving the natural upstream production of D .

Three propositions must therefore remain distinct: the natural provenance of D identified by the selection-provenance test; the local causal relation $D \rightarrow E_F$ identified under suitable variation in D ; and any source-target validity basis required to use that local relation in the target regime. Realization provenance involving E_F , relevant nonfocal variables, and Y remains separately identified.

Worked counterexample: a within-realization-test source-target validity failure. For this binary worked example, the coding is $A \equiv 1$ and $B \equiv 0$. Consider two causal systems, MV and MN. In the target regime, both satisfy the same candidate-participation and selection certificates, including the independently validated condition $D=A$. Thus the inter-stage prerequisite $H_{0,1}^{pre}$ can be satisfied in both systems.

In a separate local realization-test source regime $R_{1,s}$, both systems obey

$$E_F = D.$$

The same valid manipulation therefore identifies the same local relation $D \rightarrow E_F$ in both systems.

The systems differ in the target-regime realization architecture. In M_V , the target mechanism remains $E_F = D$ and $Y = E_F$. The source $D \rightarrow E_F$ relation is valid in the target regime and focal implementation contributes to realization. In M_N , the target mechanism is instead $E_F = 0$, while a nonfocal variable $x_1 = 1$ produces $Y = x_1$. The target observation can therefore still be $Y = A$, and the same local source-regime $D \rightarrow E_F$ effect has been correctly estimated, but that effect does not apply in the target regime.

The missing object in M_N is not necessarily $H_{0,1}^{pre}$. It is the within-stage record $H_{1,D \rightarrow E_F}^{val}$. Until that validity condition is supplied, the local realization-test effect cannot support the intended target-regime end-to-end attribution. The example therefore establishes the narrower result:

Locally correct causal identification in a source regime does not by itself license promotion of that causal proposition into the target-regime end-to-end attribution.

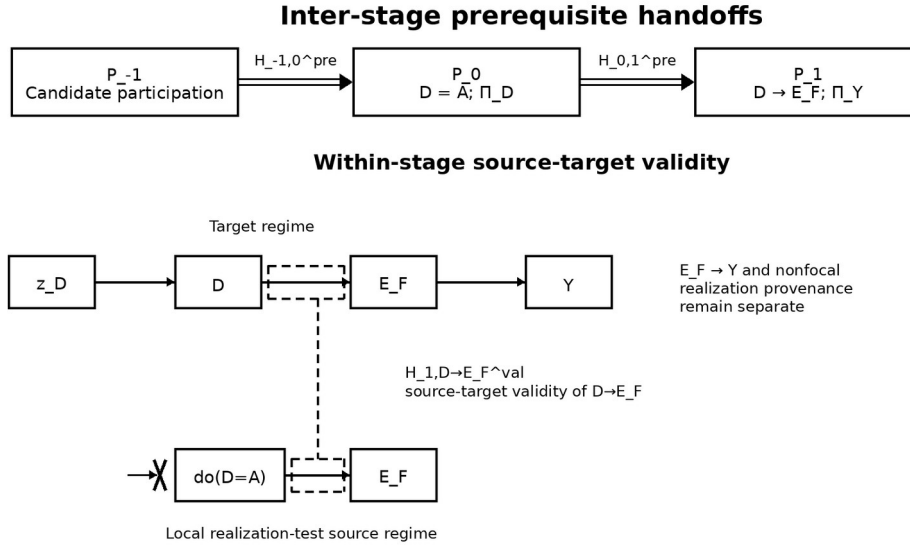


Figure 3. Inter-stage prerequisite handoffs and within-stage source-target validity. The upper row composes P_{-1} , P_0 , and P_1 using the distinct prerequisite records $H_{-1,0}^{pre}$ and $H_{0,1}^{pre}$. The lower diagrams distinguish natural selection provenance from a local realization-test source regime used to identify $D \rightarrow E_F$. A literal $do(D=A)$ severs natural incoming causes of D . The dashed $H_{1,D \rightarrow E_F}^{val}$ bridge carries only the locally identified $D \rightarrow E_F$ proposition into the target regime; it does not establish the separate realization-provenance relation $E_F \rightarrow Y$. Ordinary arrows are causal relations, double arrows are inter-stage prerequisite certifications, and the dashed bridge is a within-stage source-target validity record.

4.4 Local focal contribution versus global provenance

Even valid local composition does not establish that the focal subsystem independently originated the causally effective states in the chain.

Suppose Test 0 identifies $z_D \rightarrow D$, while opening the relevant upstream boundary also reveals $x_0 \rightarrow z_D$. Both relations can be true: z_D contributes causally to selection, while x_0 contributes causally to the state that has that effect.

Likewise, suppose Test 1 identifies $E_F \rightarrow Y$, while the realization analysis reveals $x_1 \rightarrow E_F$. Again, E_F genuinely contributes to realization while the causally effective focal implementation state has relevant nonfocal provenance.

Nothing in these upstream relations requires the local effects to be discarded. Conversely, neither local effect licenses removal of upstream provenance from the account. The resulting global constraint is therefore unchanged:

Local focal causal contributions across the identified focal relations do not imply unrestricted global focal autonomy.

This is a limit on inference from local causal results, not a general definition of autonomy.

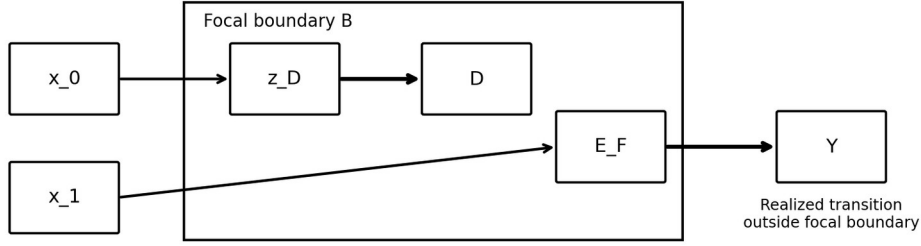


Figure 4. Local focal causal effectiveness with nonfocal provenance. The focal boundary B contains the focal selection-relevant state z_D , the selected state D , and focal implementation outcome E_F . The realized environmental transition Y lies outside the focal boundary when the target query distinguishes implementation from downstream realization. $z_D \rightarrow D$ and $E_F \rightarrow Y$ may be valid focal causal effects while $x_0 \rightarrow z_D$ and $x_1 \rightarrow E_F$ establish nonfocal preparation. Focal causal effectiveness and focal-state provenance are distinct causal facts.

5. Query-Relative Observability and Intervention Structure

The local tests define causal contrasts that are empirically usable only insofar as the roles required by the particular attribution query can be measured or causally identified. The earlier role inventory should not be read as a universal cardinal minimum. Instead,

$$\mathcal{O}(Q) = \text{the subset of } \{M_R, Z_C, z_D, X_E, D, E_F, Y, [m]\} \text{ required to distinguish the contrasts contained in } Q. \quad (10)$$

The roles are query-dependent. They may merge after causal validation; some disappear when the scientific query is narrowed; and $[m]$ is conditional on intervention-relative state insufficiency.

5.1 Query-relative observability roles

M_R is required when the query explicitly distinguishes failure of representational availability from representation without candidate participation. If that contrast is not part of the scientific claim, the framework does not require a separate M_R role merely for completeness.

Z_C is required when candidate participation must be distinguished from losing competition. The role is a validated candidate-sensitive preselection target, not a universal neural variable.

Z_D is required when the query asks whether a focal state contributes causally to D and how that contribution differs from nonfocal relay, preparation, or joint contribution.

$X_E=(x_0, x_1)$ represents the measured nonfocal variables needed for stronger provenance and external-control claims. It is not required for a purely local focal-effect claim that makes no assertion about nonfocal preparation, bypass, or reference conditions. Nor does X_E imply exhaustive enumeration of every nonfocal cause.

D is required whenever selection must be identified independently of downstream implementation or realization. Without an independent D , selection collapses into retrospective inference from outcome.

E_F is required only when the target query scientifically distinguishes focal implementation from downstream realization. If Y is defined at the focal implementation boundary, E_F and Y may validly merge and that distinction disappears. If the query keeps them separate, an independent implementation outcome remains required.

Y defines the realized transition at the scale relevant to the scientific question. Persistence may be included in the definition of Y rather than creating another generic stage.

The optional $[m]$ is retained only when otherwise matched present states respond differently under the intervention family relevant to the query. Its role is to mark present-state insufficiency, not to posit a mechanism. If an augmented present state restores the relevant intervention-relative sufficiency, raw history is not retained as an intrinsically irreducible causal variable.

5.2 Valid variable merging

Parsimony is appropriate only if it preserves the causal roles the query requires. $Z_C=Z_D$ is permitted only if one measured state independently satisfies both the candidate-participation requirement and the selection-provenance causal requirement. Correlation, common anatomy, temporal proximity, shared decoding, or prediction of one role by the other is insufficient.

The same rule applies to nonfocal variables. A single x may replace x_0, x_1 only if the same measured nonfocal state is demonstrated to supply the relevant causal role at each queried stage. $D=E_F$ is likewise permitted when one empirical state is independently validated for both the domain-specific selection criterion and the focal implementation role. $E_F=Y$ is permitted when the scientific outcome is intentionally defined at the focal implementation boundary, so that the implementation-versus-realization contrast is no longer part of the query.

Thus removing a role can have two different meanings. It may be an empirically validated merge that preserves the original query, or it may narrow the scientific query so the associated causal contrast is no longer claimed. Those cases should not be conflated.

5.3 Ordered identification dependencies and intervention families

The invariant structure is an order of logical certification, not mandatory laboratory chronology:

$$\boxed{P_{-1} \prec P_0 \prec P_1,} \quad (11)$$

where $\prec$ means that the downstream proposition presupposes the relevant upstream certificate or a validly transported equivalent. Experiments may be simultaneous, factorial, archival, sequential, or differently ordered in time. What matters is that the required causal propositions and handoffs can be established.

Test -1 requires a candidate-specific causal-identification strategy that distinguishes representational availability from candidate causal participation without confusing detection with intervention-induced recruitment. Test 0 requires causal identification of focal and relevant nonfocal contributions to D while the candidate-participation premise remains valid in the regime used for the attribution. Test 1 requires identification of $D \rightarrow E_f$ where that distinction is part of the query, plus separate identification of realization provenance involving E_f , nonfocal variables, and Y .

Intervention requirements.

Test	Conceptual intervention / identification family	What it identifies	Required validity / prerequisite condition	Failure to check	Feasibility caveat
Candidate-participation test	Candidate-specific nonconstructive perturbation or another validated candidate-specific identification strategy, with qualification/incumbent controls as required	Intervention-relative candidate causal participation	Representational criterion and independently validated Z_c remain appropriate in the tested regime	Probe recruits/reactivates the candidate, alters other candidates, or defines Z_c circularly	Selective nonconstructive perturbation may be unavailable
Selection-provenance test	Intervention or valid exogenous variation in Z_D and relevant X_0	Selection provenance of independently identified D	Candidate-participation proposition is revalidated or validly carried into the Test-0 regime	Manipulation changes the participating candidate set	Selective focal/nonfocal interventions may be unavailable
Realization-provenance test	Valid variation in D , E_f , and relevant X_i as required by the query	$D \rightarrow E_f$ and realization provenance	Prerequisite: $D=A$ remains valid or is validly carried into the realization-provenance attribution regime. Separately, any causal relation such as $D \rightarrow EF$ estimated in another regime requires its own source-target validity basis before	$D \rightarrow E_f$ changes across regimes; realization provenance is inferred rather than identified	Literal intervention on abstract selection or implementation may be unavailable

Test	Conceptual intervention / identification family	What it identifies	Required validity / prerequisite condition downstream use.	Failure to check	Feasibility caveat
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5.4 Conceptual identifiability versus current manipulability

A causal-identification protocol can state the observation, intervention, or transport condition required for a claim even when that operation is not technologically available in a particular domain. Selective perturbation of an abstract candidate, independent validation of a preselection target, measurement of focal implementation, or transport of a manipulated selection relation may each be difficult or unavailable.

Technological unavailability is not evidence that the causal distinction is unreal. Conversely, conceptual imaginability of an ideal intervention is not evidence that the corresponding attribution has been established. When a required empirical capability or inherited transport condition is unavailable, the relevant attribution remains unidentified for that domain and query.

5.5 Illustrative mapping to constrained reach selection

A constrained reaching-choice paradigm illustrates how the protocol can discipline attribution in a recognizable empirical setting. This is an **illustrative mapping of possible empirical roles; no cited study is claimed to have instantiated the complete protocol**.

Cisek and Kalaska (2005) showed that dorsal premotor population activity can simultaneously reflect multiple potential reach directions before later differentiating toward the selected reach, supporting a distinction between potential-action specification and final selection. A direction-specific component associated with alternative A could supply a candidate-specific experimental variable q_A , but decoding that component would establish representational/action-specification availability rather than candidate causal participation. A measured preparatory signal could count as Z_C only if the component used for the inference had independent causal relevance to the prespecified reach-selection attribution query in the regime where the inference is made, or a justified validity basis into that regime. Generic motor excitability would not suffice. Bestmann and Duque (2016) emphasize that action-preparation readouts and TMS effects can reflect several overlapping processes, reinforcing the need for construct-specific validation.

$D=A$ could be instantiated by an independently validated population-level selection criterion that distinguishes the selected reach before overt execution; the protocol does not require that criterion to be discrete or instantaneous. E_F could be instantiated by an independently validated target-specific descending motor or EMG outcome establishing focal implementation. Preparation and movement-generating activity need not coincide: Kaufman et al. (2014) distinguish output-null preparatory activity from output-potent movement-related activity, and Kim et al. (2021) review reaching in terms that distinguish action selection from movement implementation. The external hand or cursor transition to target A constitutes Y .

The attribution consequences are direct. Decoding A without a selection-relevance-valid candidate-specific causal manipulation licenses representation, not the candidate-participation certificate. Independently identifying $D=A$ without an implementation observable licenses selection but leaves focal implementation unresolved. A local stimulation-based estimate of $D \rightarrow E_F$ obtained under an altered network state cannot automatically be used in the natural target regime; Bergmann and Hartwigsen (2021) emphasize that stimulation-to-behavior inference contains multiple causal links, potential network propagation, state dependence, and construct-validity challenges. The stronger end-to-end attribution therefore requires the missing certificate or source-target validity basis rather than a new motor mechanism.

6. Identification Limits

The protocol distinguishes specified causal-provenance structures only to the extent supported by the measured boundary, query, intervention family, and validity assumptions of the local certificates. Three limits remain even when the protocol is applied correctly: untested upstream provenance can remain outside the model; interventional causal contribution does not automatically determine token-level necessity or sufficiency; and missing empirical capabilities leave explicit causal equivalences unresolved.

6.1 Open-boundary limitation

Suppose the tested model establishes $Z_D \rightarrow D$ and $E_F \rightarrow Y$, and the measured nonfocal variables resolve the preparation, bypass, and joint-contribution alternatives included in the query. These results support bounded focal causal contributions. They do not establish that the tested boundary contains every upstream cause of Z_D or E_F .

Let U denote an omitted upstream state outside the tested model. The actual system may contain $U \rightarrow Z_D \rightarrow D$ or $U \rightarrow E_F \rightarrow Y$ without contradicting the local focal effects. U is not a proposed mechanism; it represents the logical possibility that relevant provenance remains outside the tested boundary.

No finite tested model therefore establishes the absence of every conceivable omitted upstream cause. This does not make causal attribution impossible. Bounded causal contribution may be identifiable relative to a specified model, query, intervention family, and tested nonfocal set when the required causal assumptions, observations, interventions, and validity conditions are satisfied. Unrestricted causal origin lies outside the licensed claim.

6.2 Token-causation limitation

The local tests establish interventional causal contribution at the level supported by their design. That claim must be distinguished from necessity or sufficiency for one particular realized token.

Actual-causation theory distinguishes general causal relations from actual or token causation in a particular causal setting (Halpern, 2016). The probabilities-of-causation literature likewise

treats necessity, sufficiency, and necessity-and-sufficiency as distinct counterfactual quantities whose identification generally requires information or assumptions beyond an average or population-level interventional effect (Pearl, 1999; Li & Pearl, 2023).

Related work has quantified agents' causal responsibility for later states of dynamical systems using necessity- and sufficiency-based measures (Stecher & Baumgärtner, 2024). The present protocol addresses a different question: whether stage-resolved causal propositions are admissible for promotion into a bounded focal attribution, while explicitly declining to infer technical or token necessity or sufficiency without additional assumptions.

Consequently, evidence that intervention on Z_D changes the distribution of D , or that intervention on E_F changes the distribution of Y , does not by itself establish that the observed value was necessary for the particular token that occurred. Stronger token claims require independently supplied actual-causation assumptions or identification conditions. The present protocol does not add a token-causation test.

6.3 Measurement, intervention, and handoff limitations

When an empirical requirement is absent, the appropriate response is to state the unresolved equivalence rather than infer the missing stage from downstream behavior.

Without an independent M_R in a query that distinguishes representation from participation, absence of representation cannot be distinguished from representation without causal participation.

Without a Z_C whose relevant component has independently established causal relevance to the prespecified selection-attribution query in the inference regime, or a justified validity basis into that regime, an effect on the measured preselection signal does not certify selection-relevant candidate participation.

Without a candidate-specific intervention or equivalent validated causal-identification strategy, representation may be established while the strongest Test -1 candidate-participation certificate remains unidentified.

Without an independent D , selection cannot be distinguished from downstream implementation or realization.

Without causally identifiable Z_D or relevant x_0 , the corresponding Test-0 provenance alternatives remain unresolved.

Without an independent E_F when the query distinguishes implementation from realization, focal implementation failure cannot be distinguished from downstream realization failure, assistance, or bypass.

Without valid causal identification of $D \rightarrow E_F$ association between selection and implementation cannot supply that relation.

Without relevant nonfocal interventions or identifying variation, focal causal contribution may still be identified locally, but stronger claims excluding tested nonfocal preparation, bypass, or joint contribution cannot be made.

Without an appropriate certified validity record $H=(P, R_s, R_b, V)$, a proposition established in one regime cannot be used as a premise for the downstream attribution in another regime merely because both local analyses are individually correct.

Finally, if intervention-relevant historical dependence remains after the proposed present state is matched, that present-state description remains insufficient for the query. The appropriate response to all of these limits is to state the unresolved equivalence explicitly. **Promotion of the complete end-to-end attribution stops at the first missing required proposition or validity condition; later-stage local causal relations may still be studied and identified independently.**

7. Implications for Empirical Attribution

The protocol supports a hierarchy of focal causal attributions rather than a binary judgment about whether the focal subsystem controlled an observed transition. Stronger language requires additional causal propositions, valid handoffs, and broader tested provenance.

7.1 Hierarchy of warranted focal claims

Level 1: focal causal participation at one stage. Evidence establishes that a focal variable contributes causally to one local target, such as $Z_D \rightarrow D$ or $E_F \rightarrow Y$. The licensed claim is local. It does not imply complete end-to-end focal attribution.

Level 2: certified candidate participation with focal causal contribution across identified selection and realization stages. Test -1 establishes the candidate-participation prerequisite, while Tests 0 and 1 establish the relevant focal causal contributions at the selection and realization stages, with the required handoffs and validity conditions certified. This licenses an end-to-end attribution in which the candidate causally participated and focal causal contribution is identified across the selection and realization stages. It does not establish that candidate-stage participation itself was focal or that the focal subsystem originated the causally effective focal states.

Level 3: focal contribution surviving tested nonfocal controls. Relevant tested nonfocal preparation, bypass, and joint-contribution alternatives at the selection and realization stages represented in $X_E = (x_0, x_I)$ have been examined, and the focal contributions remain. This licenses a stronger provenance claim relative to that tested nonfocal set. It does not establish focal provenance of candidate participation or exclude untested upstream causes.

Level 4: focal realization under defined external reference conditions. All lower certificates hold; the tested nonfocal support variables are placed in prespecified reference or withdrawn conditions compatible with the scientific query; and the complete target transition remains supported or realizable under those conditions. What counts as the transition remaining supported or realizable must be specified in advance by the target scientific query and

operationalized using the domain-appropriate outcome criterion. This licenses a statement that, within boundary B and those specified external reference conditions, the identified focal causal contributions support the complete transition. It does not establish technical causal sufficiency in the counterfactual sense, token sufficiency, self-causation, metaphysical independence, or absence of every external antecedent.

Warranted claim	Minimum supporting evidence for that claim	What it still does not establish
Focal causal participation at one stage	Valid local causal identification of a focal contribution	Focal contribution elsewhere; end-to-end composition; focal origin
Certified candidate participation with focal causal contribution across identified selection and realization stages	Certified candidate participation (Test -1); identified focal selection and realization contributions (Tests 0 and 1); required prerequisite handoffs and validity conditions	Focality or provenance of candidate-stage participation; origin of focal states; exclusion of untested nonfocal provenance
Focal contribution surviving tested selection/realization-stage nonfocal controls	Level 2 plus identification of tested selection/realization-stage preparation, bypass, and joint-contribution contrasts in $X_E=(x_0,x_1)$	Focal provenance of candidate participation; absence of untested upstream causes; unrestricted autonomy
Focal realization under defined external reference conditions	Level 3 plus maintenance of the complete target transition under prespecified tested external reference conditions	Technical/token sufficiency; self-causation; absence of every external antecedent

7.2 What local contribution does not establish

The principal global inference constraint remains: **local focal causal contributions across the identified focal relations do not imply unrestricted global focal autonomy.**

Both $z_D \rightarrow D$ and $x_0 \rightarrow z_D$ may hold. Likewise, both $E_F \rightarrow Y$ and $x_1 \rightarrow E_F$ may hold. The first relation in each pair identifies what a focal state causally contributes downstream. The second identifies relevant provenance of that causally effective focal state. Focal causal effectiveness and provenance of the focal state are different attribution questions.

Consequently, a phrase such as “the focal system controlled the transition” is scientifically underspecified unless the intended evidential level is stated. It may refer to a local focal causal contribution, certified candidate participation with focal causal contribution across the identified selection and realization stages, focal contribution surviving the tested selection- and realization-stage nonfocal controls, or focal realization under specified external reference conditions. Those claims are not interchangeable.

7.3 Bounded endpoint

The central question is: **what focal causal attribution does the available evidence actually license?** For a local attribution, the answer may stop at one successfully identified stage. For the **complete end-to-end attribution query addressed by this manuscript**, stronger promotion requires candidate-participation, selection-provenance, and realization-provenance certificates; certified validity of the propositions used at their interfaces; and preservation of the tested nonfocal preparation, bypass, and joint-contribution structure.

Accordingly, an apparently agentic outcome licenses the manuscript's complete end-to-end focal attribution only to the extent that those causal propositions are independently identified and validly composed in the regimes to which the attribution applies. The claim remains bounded by the stated causal boundary, query, intervention or observation families, target outcome, and tested nonfocal variables.

The protocol therefore converts an apparently unitary attribution into explicit stage-resolved causal-identification requirements. Its scientific content lies not in new causal machinery, but in specifying which inherited causal query must be answered at each attribution stage, what proposition must remain valid at each interface, and the precise point at which stronger end-to-end focal attribution becomes unidentified.

Generative-AI Disclosure

OpenAI ChatGPT was used as a research-assistance tool during the 2026 research and manuscript-development process, including conceptual formalization; literature discovery and organization; adversarial critique and stress-testing; drafting and editing; mathematical and causal notation refinement; simulated peer-review workflows; and manuscript and submission-package preparation. Multiple model versions available through the ChatGPT service were used across the project; GPT-5.6 Sol was used during final submission-package preparation. The author originated and directed the research questions and analytical program, adjudicated model outputs, determined which claims were retained, independently reviewed the scientific argument, and verified cited sources and relevant factual claims. AI-assisted critique informed revision and presentation, but the author retained intellectual control over the framework and conclusions and accepts full responsibility for the submitted manuscript. Generative AI is not an author and generated no empirical data.

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